

BCG: A Game-Based Approach to Motor Imagery Adaptation and BCI Calibration

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1. Introduction

What is EEG-based Motor Imagery?

Electroencephalography-based motor imagery (EEG-MI) is a technology that reads brain signals when a person imagines moving, allowing those signals to be translated into commands for electronic devices (Wolpaw et al., 2002). It can be used to help people control wheelchairs, robotic arms, and computer interfaces.

However, these brain signals vary from person to person and can also change over time (Lotte et al., 2018), so each MI session usually needs a calibration step before it can work properly.



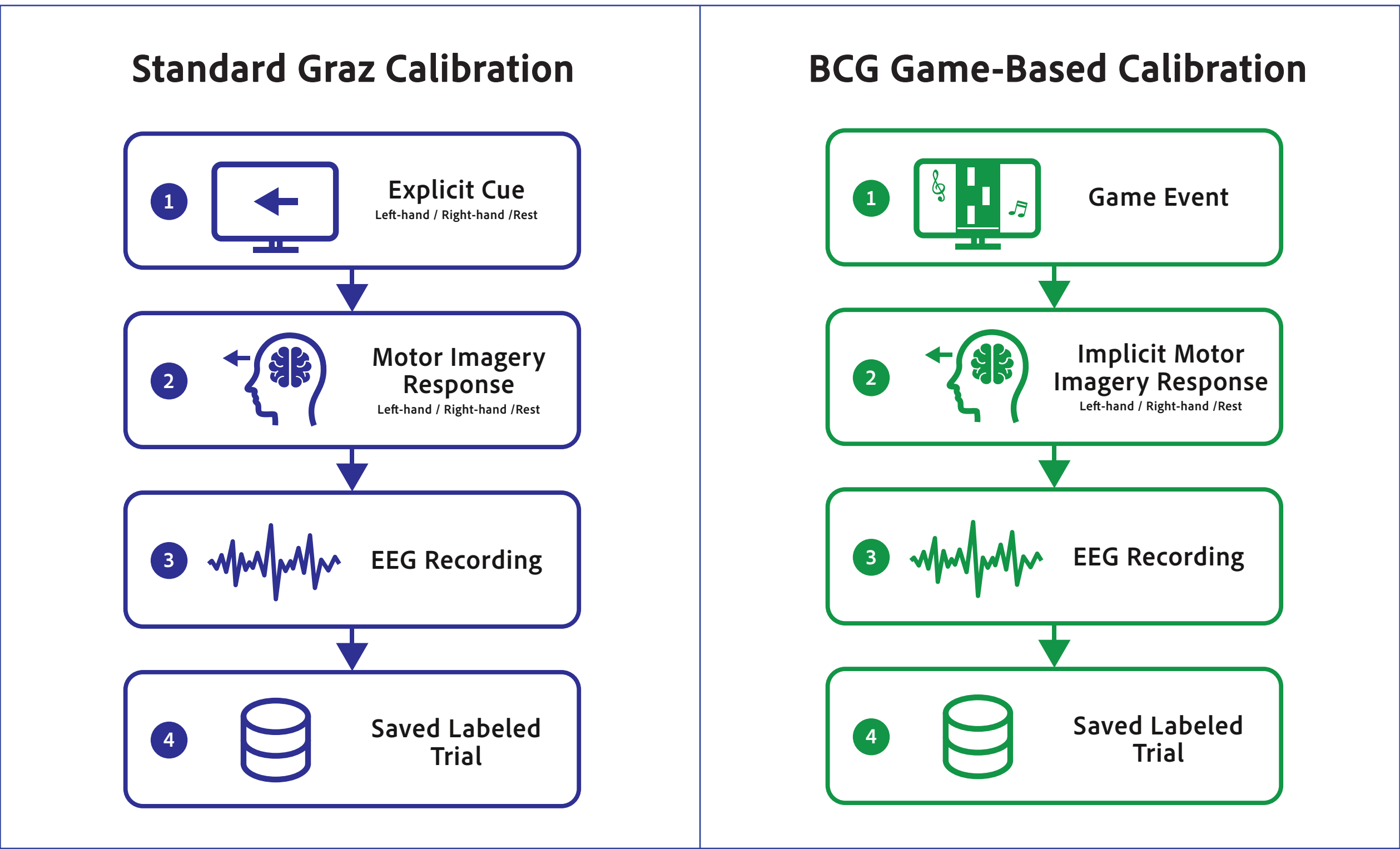
Source: The University of Texas at Austin News

The Traditional Calibration Method

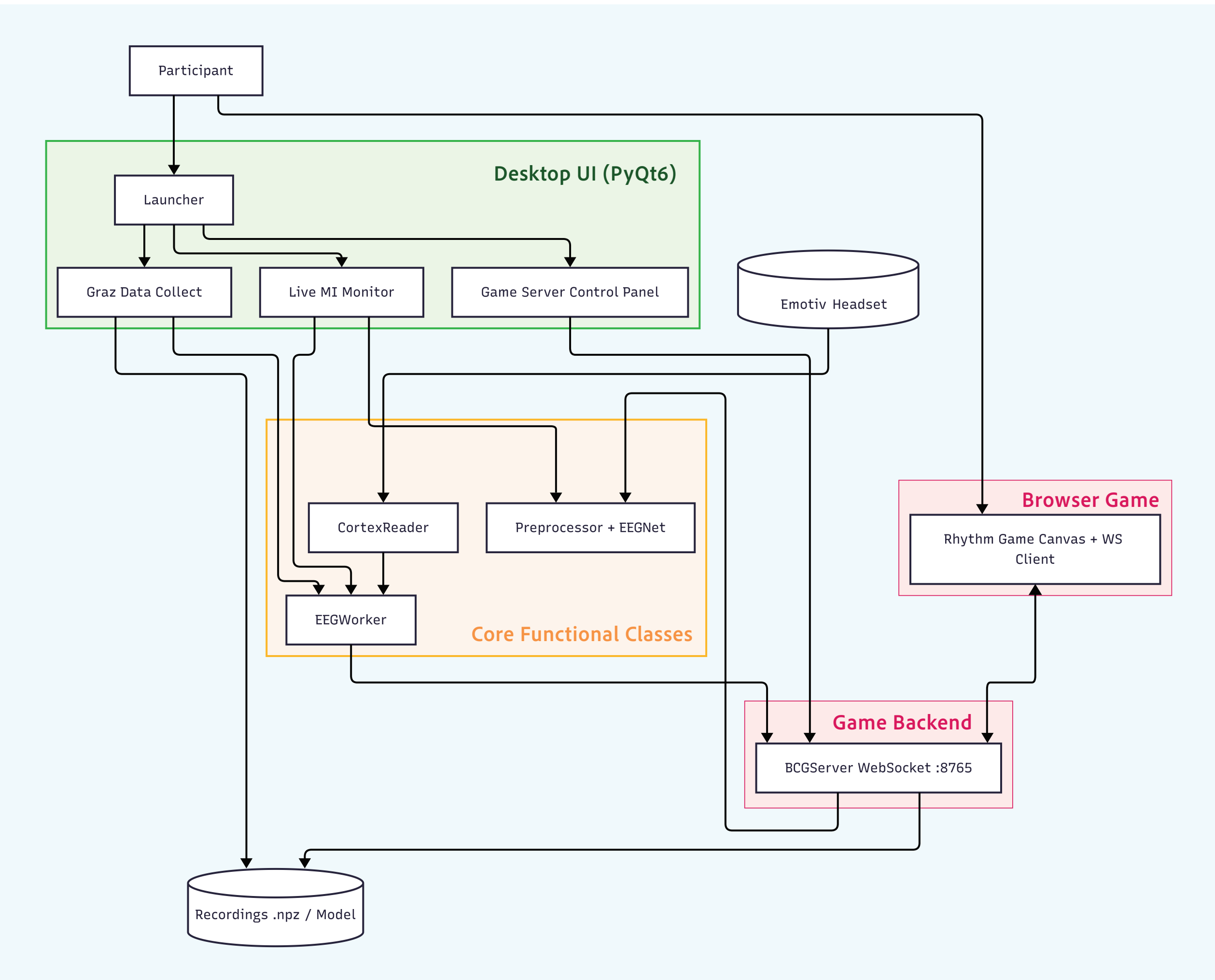
Calibration is the process where the system collects brain signals from the user while they perform specific tasks, then learns from those signals. The Graz method asks users to follow repeated on-screen prompts and imagine the same movement many times during each session. This method is still effective, but it usually takes 15–30 minutes per session, requires expert supervision, and can be tiring (Giles et al., 2022). As a result, it may reduce user engagement and lower data quality over time (Lotte et al., 2018).

2. Methodology

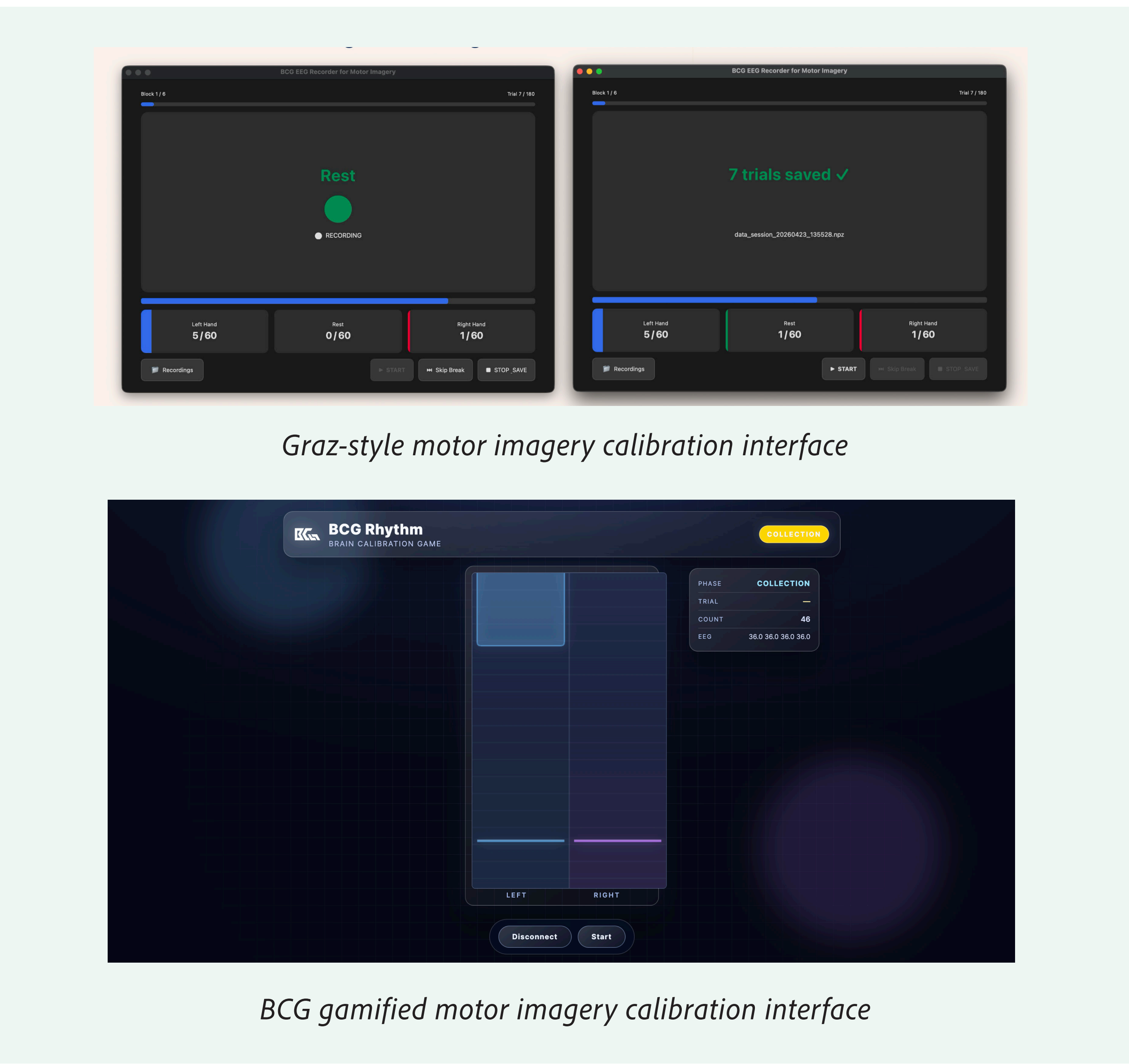
BCG Concept Compared to Graz Method



System Architecture



BCG Interface Compared to Graz Interface

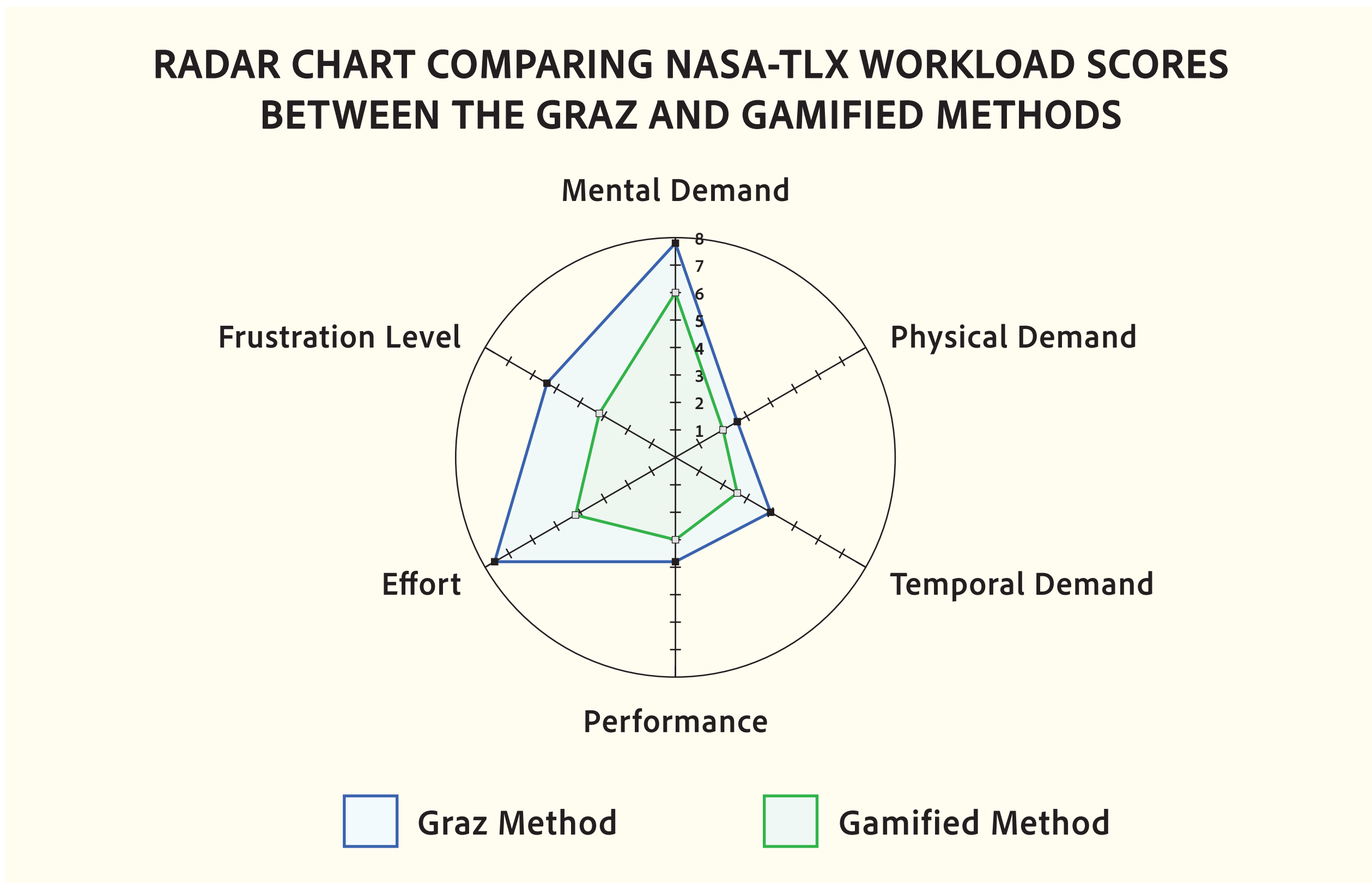


3. Evaluation

Evaluation Design

Between-subjects user study with 10 healthy participants to compare the Graz protocol and the gamified protocol. Participants were split into two groups, with 5 people in each condition. Each participant completed a 10-minute calibration task, and we used NASA-TLX to measure perceived workload and user experience.

User Study Result



4. Conclusion

- This study explored a gamified approach to BCI calibration and its impact on user workload and experience.
- The user study with 10 participants suggests that the gamified interface may reduce perceived effort, frustration, and mental demand compared with the traditional Graz method.
- Overall, the findings indicate that gamification could help make motor imagery calibration more engaging and practical, supporting broader real-world adoption of motor imagery-based BCI systems.

References

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Lotte, F., Bougrain, L., Cichocki, A., Clerc, M., Congedo, M., Rakotomamonjy, A. and Yger, F. (2018) 'A review of classification algorithms for EEG-based brain-computer interfaces: a 10 year update', *Journal of Neural Engineering*, 15(3), p. 031005.

Wolpaw, J., Birbaumer, N., McFarland, D., Pfurtscheller, G. and Vaughan, T. (2002) 'Brain-computer interfaces for communication and control', *Clinical Neurophysiology*, 113(6), pp. 767–791.